

Data Preprocessing & Cleaning Report

This report shows dataset overview, missing values handling, and imputation techniques (Mean, Median, KNN, MICE).

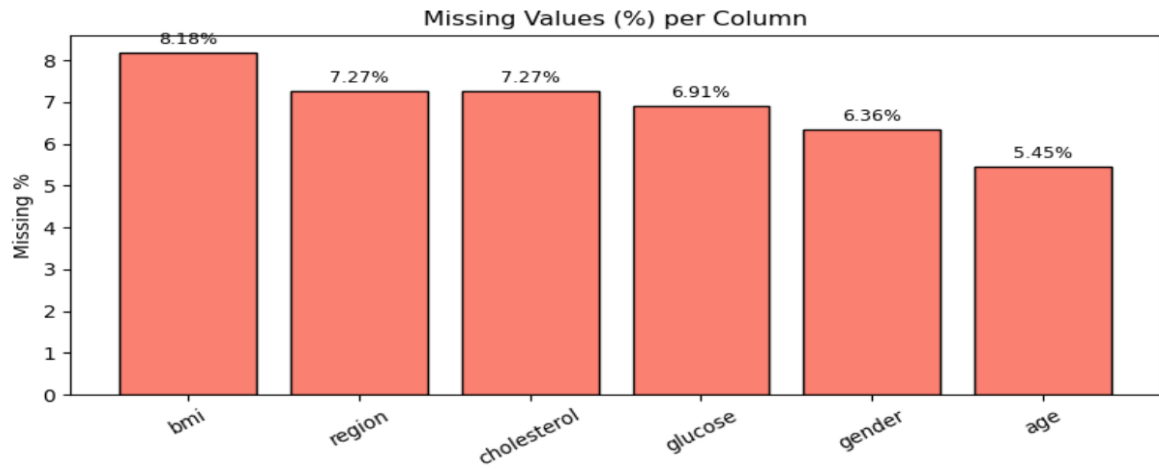
Dataset shape: (550, 9)

	patient_id	age	gender	region	bmi	blood_pressure	cholesterol	glucose	disease_risk
0	P0001	58.0	Female	East	30.04	95.9	179.2	71.3	0
1	P0002	71.0	Female	West	22.49	112.0	206.8	83.0	1
2	P0003	48.0	Female	North	21.72	110.0	204.3	90.5	1
3	P0004	34.0	Male	West	27.12	131.7	199.0	67.9	1
4	P0005	62.0	Female	South	25.34	104.8	163.2	NaN	1
5	P0006	27.0	Male	North	30.67	108.1	NaN	106.1	0
6	P0007	80.0	Female	South	28.69	113.0	257.2	119.3	0
7	P0008	40.0	Male	West	29.00	110.5	195.2	87.9	0
8	P0009	58.0	Female	West	NaN	135.8	165.7	107.9	0
9	P0010	77.0	NaN	South	NaN	103.2	NaN	62.5	1

```
: print(df.info())
print()
```

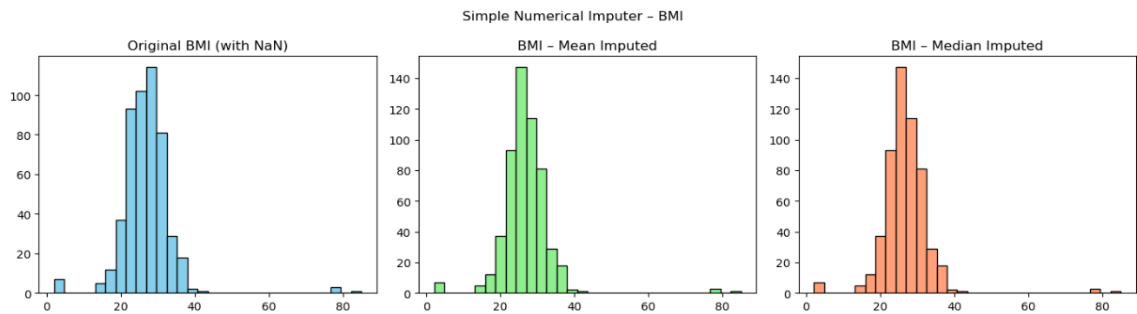
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550 entries, 0 to 549
Data columns (total 9 columns):
#      Column              Non-Null Count  Dtype
---  -
0     patient_id            550 non-null   object
1     age                   520 non-null   float64
2     gender                515 non-null   object
3     region                510 non-null   object
4     bmi                   505 non-null   float64
5     blood_pressure        550 non-null   float64
6     cholesterol           510 non-null   float64
7     glucose               512 non-null   float64
8     disease_risk          550 non-null   int64
dtypes: float64(5), int64(1), object(3)
memory usage: 38.8+ KB
..
```

	MISSING COUNT	MISSING %
bmi	45	8.18
region	40	7.27
cholesterol	40	7.27
glucose	38	6.91
gender	35	6.36
age	30	5.45

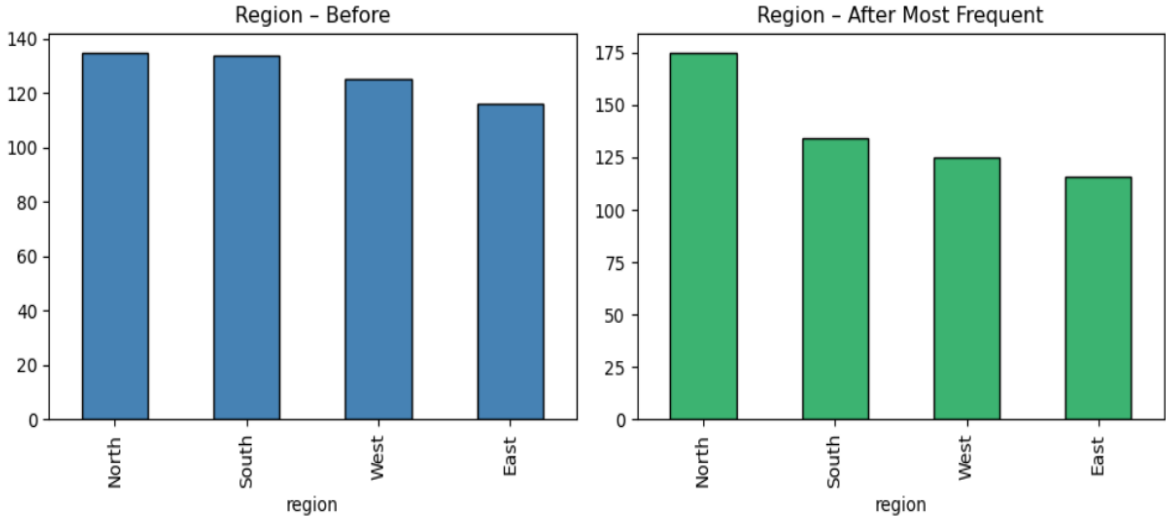


Rows where BMI was missing (first 10):

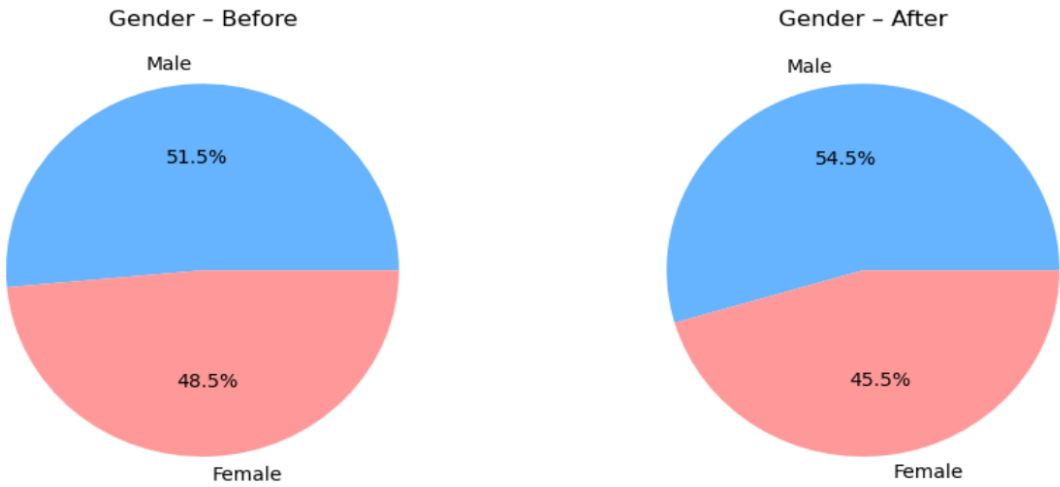
	patient_id	bmi	bmi_mean	bmi_median
8	P0009	NaN	26.91804	26.85
9	P0010	NaN	26.91804	26.85
23	P0024	NaN	26.91804	26.85
33	P0034	NaN	26.91804	26.85
52	P0053	NaN	26.91804	26.85
57	P0058	NaN	26.91804	26.85
66	P0067	NaN	26.91804	26.85
77	P0078	NaN	26.91804	26.85
104	P0105	NaN	26.91804	26.85
130	P0131	NaN	26.91804	26.85



Most frequent region used: North
Missing before: 40
Missing after: 0

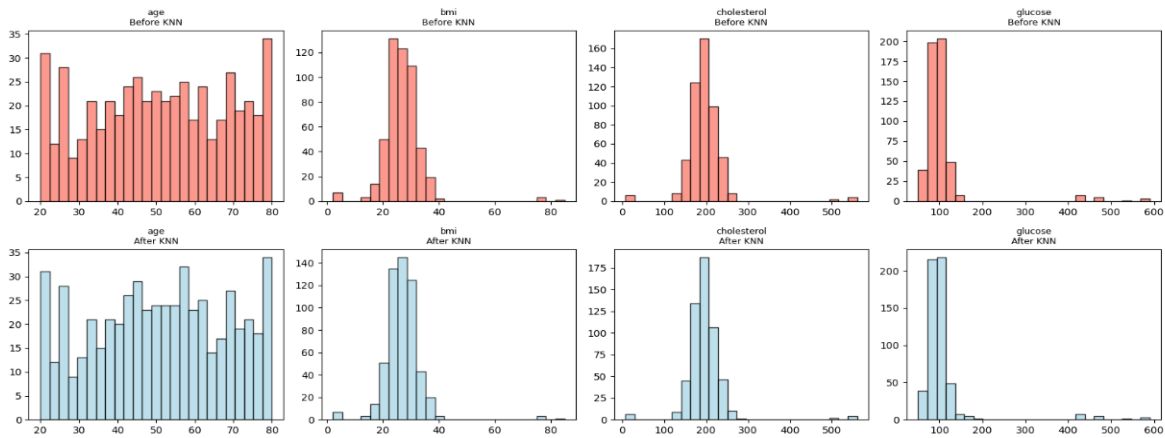


Most frequent gender: Male
Missing before: 35
Missing after: 0



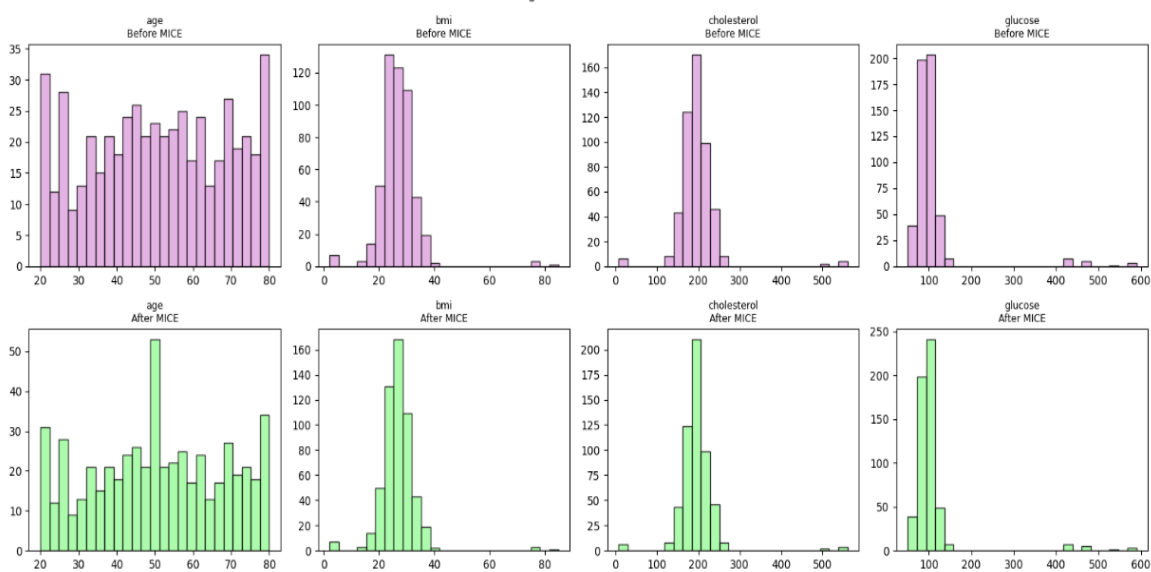
Missing after KNN:
age 0
bmi 0
cholesterol 0
glucose 0
dtype: int64

KNN Imputer - Before vs After



75%	65.00	29.54	211.65	107.02
max	80.00	85.00	560.00	590.00

MICE Algorithm - Before vs After



Correlation Heatmap - Final Clean Dataset

